

EC441 Midterm Two Spring 2022

Name 1: _____ Name 2: _____ Version 406

This is a closed book exam. You may use one 8.5x11 piece of paper, typed or written on both sides, that can be read normally (no micro printing). Calculators are allowed. The time for the exam is 90 minutes.

1 TCP (20 points)

- (a) If you connect a raspberry-PI to a fast desktop computer using a 1 Gb/s Ethernet link, and attempt to send a large file from the desktop to the PI, will TCP congestion control or TCP flow control be the more important mechanism to make this work properly? Explain.
- (b) Suppose the round trip time between a web client and a web server is 5 ms. What is the usual delay before a small web page can be displayed by the web client, starting from the time the URL is entered into the browser by the user? Assume that TCP is used and that this is the first request from this client to this server.
- (c) Suppose you are sending packets with 1000 bytes of application layer data from A to B. A receives a packet with the ACK number of 12345 and a sequence (SEQ) number of 54321. What would the next packet that A receives be ? (i.e. what would the ACK and SEQ numbers be?) If there are several likely options, explain.

2 Networks and Hosts (30 points)

- (a) Which of the following are valid IPv4 networks? If it is a valid network, what is the CIDR representation (i.e. a.b.c.d/x)?

Net Address	Net Mask
10.0.0.0	255.0.0.0
192.168.0.0	255.255.255.0
128.197.128.0	255.255.192.0
128.197.128.9	255.255.192.0
128.197.128.40	255.255.255.248

- (b) Someone tells you about their IPv4 network over the phone, but you only hear this:

187.45.____6.0/19

You are missing zero, one, or two digits where the blank is: the third number could be 6, 86, 186, 986, etc.

What is the correct value for the third number?

- (c) You are an ISP and have been assigned the network 191.45.12.0/22. You wish to break this into /24 and /25 networks. Provide an example of a decomposition of the original network that uses all the addresses and includes both /24 and /25 networks.

3 Error Control (25 points)

- Design an error control code with 2 data bits that can be used for single bit error detection. Use the fewest number of redundancy bits as possible. Your answer should be in the form of a set of n-bit codewords, where your task is to determine both n and the codewords.
- Design an error control code with 2 data bits that can be used for single bit error correction. Use the fewest number of redundancy bits as possible. Your answer should be in the form of a set of n-bit codewords, where your task is to determine both n and the codewords.
- What is the minimum distance achieved by the IP checksum coding scheme? Could IP do error correction using its current checksum structure?

4 Routing and AS (25 points)

- Consider a network with 24-bit addresses for the hosts. The maximum size of a subnet is 256 hosts. The routing table entries are of the form a.b.c/x (dotted decimal with CIDR notation). This means that all the hosts that are inside a subnet are represented by a single routing table entry.

You would like the routing tables of all routers (inter-AS routers, intra-AS routers, gateway routers) to be no larger than 500 entries.

How can this be accomplished? How would you break the network into autonomous systems? What should the size of the ASes be?

- Find the shortest path between node A and node g in the following network. Describe the path like this: ABcZg. What is the cost of this path?
- If the link cost from g to h changed from 3 to 1, would the path from the previous question change? If so, what would the new path be?

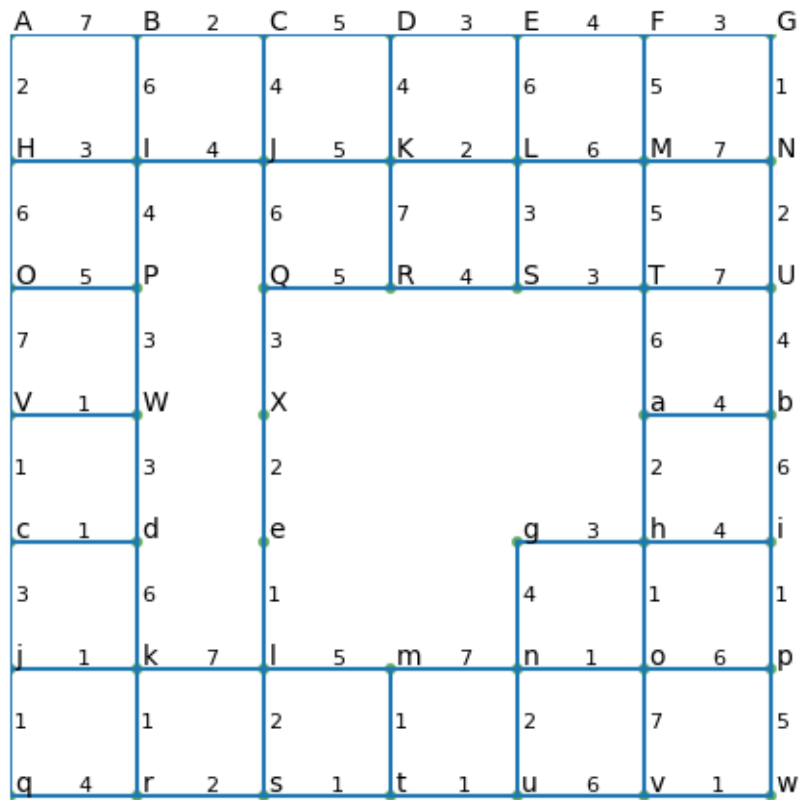


Figure 1: Network with costs indicated for Problem 4